Supplementary File 10: k-means cluster centroids

Shrinkage Level	Level 1	Level 2	Level 3	Level 4
DE_Lüneburg Region_Variable	Cluster 4	Cluster 3	Cluster 1	Cluster 2
CLOS_CENT	0.5095	0.44	0.566	0.6426
DEGREE	4	3	4.5	5.75
ECCENTRICI	3.5	4	3	2.75
EIGEN_CENT	0.3804	0.4954	0.7016	0.8189
ELDER_RATE	26.55	25.3	22.625	21.475
FAC_COMM	387	1253	737.75	978.75
FAC_SOC	211	525	361.75	542
FAC_TRANS	18.5	40	35.25	54.75
FOR_HA	49919.514	18799.1085	26007.3556	52235.7641
POP_CHG	-3.3721	-1.5849	1.4731	3.0115
UNEMPLO_RA	6.45	5.2	4.775	4.7
URBAN_HA	3792.0517	9652.4953	7181.9161	9317.3392
WEIG_DEG	22	20	21.5	25.75

Shrinkage Level	Level 1	Level 2	Level 3	Level 4
TR_Trakya Region_Variable	Cluster 2	Cluster 3	Cluster 4	Cluster 1
DEGREE	2.8333	5.2727	6.5	6
DEV_IND	0.3415	0.4872	2.3856	1.8642
DIST_TEKIR	115.3991	76.1986	85.8898	0
EIGEN_CENT	0.2149	0.4794	0.5856	0.4315
ELDER_RATE	20.5333	13.5309	7.5875	10.3
FAC_COMM	13.75	98.8182	176	218
FAC_REC_AT	7.1667	24.5455	65.25	179
FAC_SOC	5.1667	27	65.25	70
FAC_TRANS	1	14.6364	42.25	78
INDUST_HA	86.9468	609.7569	1462.5697	819.7843
POP_CHG	-8.8594	-0.8385	12.5098	18.4246
URBAN_HA	724.2966	1802.1957	1950.9777	3587.5799
WEIG_DEG	8.125	18.8182	29.75	24.5

Shrinkage Level	Level 1	Level 2	Level 3	Level 4
DE_Lüneburg District_Variable	Cluster 4	Cluster 3	Cluster 2	Cluster 1
AGRI_HA_	17929.9321	5725.7758	3373.0734	2985.4438
BETW_CENT	33.582	472.7941	10.9158	61.3023
DIST_TO_CE	34.4	0	16.4348	11.3571
ECCENTRICI	5.6	3	5.1739	4
ELDER_RATE	23.52	18.1	17.9043	16.9071
FAC_COMM	24.6	536	6.6957	12.2143
FAC_SOC	15	226	4.8696	10.9286
FAC_TRANS	60.4	367	15	18.0714
INDUST_HA	19.4975	2359.2016	6.4992	20.8902
POP_CHG	-2.2307	6.7214	8.6126	9.678
URBAN_HA	685.0134	5764.7708	281.7252	432.1953
WEIG_DEGR	12.5	53.5	6.8696	13.1429

Shrinkage Level	Level 1	Level 2	Level 3	Level 4
DE_Lüchow-Dannenberg_Variable	Cluster 4	Cluster 3	Cluster 2	Cluster 1
BETW_CENT	21.4606	5.8595	8.0035	84.4889
DISTANCE_LUCHOW	13.495	25.7477	17.8009	8.1138
ECCENTRICI	4.0769	4.7143	5.2	3
ELDER_RATE	22.2538	29.8429	26.7	24.45
FAC_ATT	8.3077	19.5714	11	41
FAC_COMM	4.3077	12.5714	7.6	90
FAC_SOC	3.6154	5	7.8	41.5
FAC_TRANS	9.6154	8.1429	21.2	27
INDUST_HA	55182.4566	0	0	2421130.282
POP_CHG	-0.4112	-0.5191	-0.3597	-0.1822
URBAN_HA	194.5182	322.8174	152.4415	1309.2038
WEIG_DEGR	10.7692	7	7.5	26.25
WETWAT_HA	25.5219	385.5874	0	165.8923

Shrinkage Level	Level 1	Level 2	Level 3	Level 4
TR_Lüleburgaz_Variable	Cluster 1	Cluster 4	Cluster 3	Cluster 2
BETW_CENT	11.744	60.1823	22.5951	258.296
CLOS_CENT	0.295	0.38	0.3626	0.4861
DEGREE	2.9474	5.75	4	9
DIST_TO_LU	17.4358	12.9508	14.325	0
ECCENTRICI	6	4.8333	4.75	4
FAC_COMM	0.3684	0.3333	10.5	40
FAC_REC_AT	0.1053	0.9167	6	18
FAC_SOC	0.8947	1	7	65
FAC_TRANS	0.1579	0.1667	0.25	16
FOREST_HA	308.0376	48.5048	30.3527	0
INDUST_HA	16.163	75.8817	244.607	871.3009
POP_CHG	-25.2626	-16.8806	5.1968	19.9806
URBAN_HA	73.1096	114.8641	164.2617	1196.2533
WEIG_DEG	3.6053	12	6.625	21

Shrinkage Level	Level 1	Level 2	Level 3	Level 4
TR_Babaeski_Variable	Cluster 4	Cluster 3	Cluster 1	Cluster 2
BETW_CENT	14.012	50.3172	64.0222	255.5611
CLOS_CENT	0.2656	0.3451	0.3338	0.443
DIST_TO_CE	16.1748	9.1345	12.0082	0
ECCENTRICI	6.3529	4.9231	5.4	4
ELDER_RATE	17.8393	15.2242	13.0628	6.9134
FAC_COMM	0.0588	0	0	21
FAC_SOC	0.0588	0	0.8	26
FAC_TRANS	0.0588	0	0.2	12
INDUST_HA	19.9309	6.3619	82.7053	421.5301
POP_CHG	-25.9286	-24.3634	-11.205	8.0965
URBAN_HA	81.1359	78.3956	80.0374	1024.4781
WEIG__DEG	3.4706	6.4231	14.3	19

Supplementary File 10: Settlement lists grouped by k-means clusters

Table S 10. a *Lüneburg Region settlement grouped by k-means clusters*

Negative Cases: (n:2)	Vulnerable (n:1)	Potentials (n:4)	Positive Cases (n:4)
1. Uelzen	1. Cuxhaven	1. Stade	1. Winsen
2. Lüchow-Dannenberg		2. Celle	2. Lüneburg
		3. Verden	3. Soltau
		4. Osterhole	4. Rotenburg
Percentage: %18	**%9**	**%36**	**%36**

Table S 10. b *Trakya Region districts grouped by k-means clusters*

Negative Cases (n: 11)		Vulnarable (n:11)		Potentials (n:4)	Positive Cases (n:1)
1. Demirköy	2. Ipsala	1. Kırklareli	2. Kesan	1. Edirne	Süleymanpaşa
3. Pınarhisar	4. Meriç	3. Havsa	4. Muratlı	2. Lüleburgaz	
5. Vize	6. Suloğlu	5. Babaeski	6. Saray	3. Çorlu	
7. Marmaraereğlisi	8. Lalapaşa	7. Uzunköprü	8. Kapaklı	4. Cerkezköy	
9. Şarköy	10. Kofcaz	9. Hayrabolu	10. Ergene		
11. Enez		11. Malkara			
Percentage: %40		**%40**		**%14**	**%3.70**

Table S10. c *Lüneburg settlements grouped by k-means clusters*

Negative Cases (n:5)	Vulnerable Cases (n:1)	Potential Cases (n:23)			Positive Cases (n:14)	
AmtNeuhaus	Luneburg	Adendorf	Thomasburg	Rullstorf	Reppenstedt	Embsen
Bleckede		Amelinghausen	Reinstorf	Artlenburg	Bardowick	Barendorf
Dahlenburg		Kirchgellersen	Betzendorf	Hohnstorf(Elbe)	DeutschEvern	Radbruch
Nahrendorf		Vögelsen	Oldendorf(Luhe)	Sudergellersen	Scharnebeck	WendischEvern
Tosterglope		Handorf	Hittbergen	Wittorf Mechtersen	Brietlingen	Soderstorf
		Barum	Rehlingen	Ludersburg	Melbeck	Echem
		Westergellersen	Barnstedt		Neetze	Vastorf
		Dahlem	Boitze			
%15.50	**%2.20**	**%51.11**			**%31**	

Table 10. d *Lüchow-Dannenberg settlements grouped by k-means clusters*

Negative Cases (n:7)	Vulnarable (n:13)		Potentials (n:5)	Positive Cases (n:29)
1.Damnatz	1. Prezelle	2. Luckau (Wendland)	1. Lemgow	Luchow (Wendland) Dannenberg (Elbe)
2.Gartow	4. Trebel	3. Lubbow	2. Schnega	
1.Hitzacker (Elbe)	6. Gorleben	5. Woltersdorf	3. Clenze	
2.Hohbeck	9. Gusborn	7. Bergen an der Dumme	4. Waddeweitz	
3.Langendorf	10. Jameln	8. Wustrow (Wendland)	5. Zernien	
4.Neu Darchau	11. Karwitz			
5.Schnackenburg	12. Gohrde			
	13. Kusten			
Percentage: %25	**%48**		**%18**	**%7**

Table 10. e *Lüleburgaz settlements grouped by k-means clusters*

1-Negative Cases (n:17)		2- Vulnareable (n:11)		3- Positive Contrast (n:5)	4-Positive Cases (n:1)
1. Cesmekolu	2. Hamitabat	1. Tatarköy	2. Akcaköy	1. Ahmetbey	1. Lüleburgaz
3. Celaliye	4. Kırıkköy	3. Turgutbey	4. Kayabeyli	2. Ayvalı	
5. Hamzabey	6. Oklalı	5. Eskitaşlı	6. Karamusul	3. Eskibedir	
7. Ceylanköy	8. Düğüncübaşı	7. Sakızköy	8. Çiftlikköy	4. Büyükkaştıran	
9. Karaağaç	10. Alacaoğlu	9. Yenibedir	10. Saricaali	5. Durak	
11. Ertuğrul	12. Çengerli	11. Evrensekiz			
13. Emirali	14. Davutlu				
15. Umurca	16. Seyitler				
17. Yenitaşlı					
Percantage: %50		**%32**		**%14**	**%2.90**

Table 10. f *Babaeski settlements grouped by k-means clusters*

Negative Cases (n: 17)		Negative Contrast (n:13)		Positive Contrast (n:5)	Positive Cases (n:1)
1. Hazinedar	2. Müsellim	1. Kuleli	2. Pancarköy	1. Demirkapı	1. Babaeski
3. Karahalil	4. Kumrular	3. Mutlu	4. Nadirli	2. Karacaoğlan	
5. Yenimahalle	6. Düğüncülü	5. Terzili	6. Çiğdemli	3. Taşağıl	
7. Kuzuçardağı	8. Sinanlı	7. Oruçlu	8. Minnetler	4. Alpullu	
9. Nacak	10. Büyükmandıra	9. Kadıköy	10. Ertuğrulköy	5. Taşköprü	
11. Yeniköy	12. Katranca	11. Karamesutlu			
13. Karabayir	14. Çengerli	12. Osmaniye			
15. Erikleryurdu	16. Ağayeri	13. Sofuhalil			
17. Çavuşköy					
Percantage : %47.20		**%36.10**		**%13.80**	**%2.70**